colonise in the gut and then produces toxins that damage host cells.

In case of food intoxication the toxins produced by bacteria in the food are ingested. Infection and intoxication differ in their onset of symptoms. Infections are characterized by a delay in the appearance of gastrointestinal disturbance until the pathogen increases in number or affects invaded tissue. Infection is correlated with onset of fever, one of the basic body's general responses to an infective organism. In case of intoxication, the symptoms are characterized by sudden appearance of gastrointestinal disturbances like cramping, nausea, vomiting or diarrhoea.

HOTS

What is likely to happen to a child who drinks contaminated water?

12.4.1 Microbial Flora of Gastrointestinal Tract

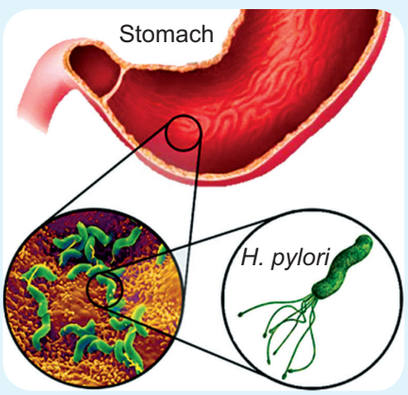
The stomach and gastrointestinal tract are not sterile and are colonized by the organisms that perform functions beneficial to the host, including the manufacture of essential vitamins. *Escherichia coli* found in the intestine help the body to produce vitamin K and *Bifodobacteria* can synthesizes vitamins such as vitamin B12, folate, and riboflavin. Humans cannot produce these vitamins. The normal flora changes according to the diet, age, cultural conditions and the use of antibiotics (Table 12.4).

12.4.2 Terms used in GIT Infections

Gastroenteritis: Inflammation of lining of stomach and intestine. It is a syndrome characterized by nausea, vomiting, diarrhea, abdominal discomfort.

DO YOU KNOW?

Stomach is acidic because of the presence of hydrochloric acid. So in this acidic condition organisms generally not survive except one bacterium *Helicobacter pylori.* This bacterium is the leading cause of stomach ulcers. This bacterium has maximum evidence of correlation with the development of stomach and intestinal cancer.



Diarrhea: Condition in which feces, are discharged from the bowels frequently and in a liquid form.

DO YOU KNOW?

Botulism is a special case of intoxication because, the ingestion of the preformed toxin affects the nervous system rather than GIT.

Infant Botulism is the infectious form of Botulism which results when spores of *Clostridium botulinum* swallowed colonise in the intestine. Botulism spores can be found in honey.

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